POL 529 - Program Evaluation Proposal

Closing the Gap for Rural Students on Postsecondary Education

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Problem Statement

High school students in rural areas often have less access to resources and advising regarding postsecondary education options than those in other areas (Lopez & Schwartz, 2019). Whiteside (2020) found this to be true when interviewing first-generation rural students, citing limited access to college information as a barrier to educational goals. The Institute of Education Sciences (2024) additionally echoes this sentiment: "Students in rural areas … often face unique constraints in access to school resources and postsecondary education." Limited information on colleges, constraints in school resources, and lack of access to sufficient individualized postsecondary education advising can place rural high school students at a disadvantage. This information gap includes knowledge of postsecondary paths, admission requirements and processes, and financial aid options such as the FAFSA (Lopez & Schwartz, 2019). Rural students are also less likely than their urban and suburban peers to have the resources needed to apply to and succeed in college, such as high-speed internet, rigorous courses, extracurricular opportunities, and fully staffed counseling offices (Lopez & Schwartz, 2019). Counselor capacity is a central part of this gap: the average student-to-counselor ratio in rural schools is 310:1 (Showalter et al., 2023). Together, limited information and limited advising can restrict students' awareness of available pathways. Nearly 10 million students attend rural public schools (McNamee et al., 2026), making access to postsecondary advising an important issue for rural education. How can we close the resource gap on postsecondary education for rural students?

Program/Policy Description

This proposal evaluates an AI-supported advising program for students in grades 9-12 at eligible rural high schools across the United States, with the goal of informing potential national implementation. The AI advising tool is the core of the program, supported by FAFSA education sessions, virtual campus tours, and postsecondary education information sessions. The program would begin with an initial group of eligible rural schools across the United States, with participating schools randomly assigned to different phases of implementation. Integrating this program during school hours beginning in 9th grade gives students more time to explore and evaluate options and determine their ideal path without pressure from admission deadlines (Kundel, 2026). Students will interact directly with the AI advising tool, which generates individualized postsecondary options from the academic achievements, hobbies, interests, and goals they provide. Students can use the tool to compare postsecondary pathways, explore admission requirements, identify financial aid opportunities, and develop individualized plans. These plans can be revisited and updated as students progress through high school. A designated counselor at each school will oversee the program, review the AI-generated recommendations for accuracy quarterly, and provide additional guidance when students have questions or need help aligning postsecondary options with their academic progress and goals. By handling routine advising questions, the AI tool may allow counselors to focus on more individualized, higher-level support (Stumpf, 2024). The program aims to increase rural students' postsecondary knowledge and confidence through AI-supported advising with counselor oversight.

Theory of Change

Problem Statement
High school students in rural areas often have less access to resources and advising regarding postsecondary education options than those in other areas. The resource gap in what is available and advice around postsecondary education can restrict students' awareness of available pathways. How can we close the resource gap on postsecondary education for rural students?

Resources	**Activities**	**Outputs**	**Short-term Outcomes**	**Mid-term Outcomes**	**Long-term Outcomes**
Computers Counselors Library Internet access AI mentor tool Students Parents Community Members	FAFSA application education sessions Attending "virtual" campus tours Attending "virtual" information sessions Interactions with AI mentor tool	Students enrolled in the AI Advisory Program Virtual Campus tours completed Students attend FAFSA education sessions AI tool usage Individualized postsecondary education planning	Increase confidence in navigating postsecondary options Increased knowledge of postsecondary options Increased financial aid (FAFSA) knowledge	Increased postsecondary applications aligned with students' goals Increased postsecondary enrollment Increased FAFSA completions	Increased degree obtainment Increased economic prosperity for individuals from a rural background Increased positive view on postsecondary education in the community

Additional considerations
Locally held beliefs regarding postsecondary education and ability to advance from a cycle of poverty. Desire of some students to not leave rural areas. Transportation cost barriers to educational facilities.

Causal Identification Strategy

Likely sources of selection bias include self-selection. Students who voluntarily participate may already be more motivated toward further education or have greater external support. Likewise, districts adopting the program voluntarily may have more resources or stronger administrative support. Comparing voluntary participants with nonparticipants could in effect produce a faulty comparison and overestimate the program's effectiveness (Gertler et al., 2016, p. 58).

To provide the program fairly while constructing a valid comparison, I will use a cluster-level randomized, phased-in evaluation, with eligible participating schools randomized by lottery to yearly implementation periods. Schools remain in the program once assigned. A preliminary sample of 60-90 schools (20-30 per stratum) will be randomly assigned within each rural classification to three annual implementation cohorts, with final allocation determined by a power calculation and available funding. Schools awaiting implementation will serve as the comparison group, with outcomes measured before they receive the program. This method serves as a valid comparison group (Gertler et al., 2016, p. 75), and it is practical because most programs must be phased in over time due to financial constraints or logistical and administrative limitations (Gertler et al., 2016, p. 191). By combining the rollout plan and lottery, this helps to prevent bias in manual selection regarding which schools receive the program before other schools. School-level randomization also reduces spillovers between classmates.

Schools must meet three eligibility criteria to participate: 1) NCES Rural Fringe, Distant, or Remote classification (NCES, 2026), 2) a graduation rate below 80%, and 3) a College Readiness Index of 50/100 or below (U.S. News, 2026). These criteria target rural schools with greater educational needs. Baseline student data will be collected by the evaluation team through an in-school survey and will include current postsecondary plans, prior school counselor engagement, and both a confidence and knowledge score regarding postsecondary options. Access to schools for survey administration will be arranged through partnerships with state education agencies and district superintendents. Outcomes will be measured yearly in both treatment and comparison schools, with the final measurement taking place one month before graduation. Postsecondary applications and enrollment outcomes will be included in post-treatment data collection.

Limitations of this program include participation, attrition, and noncompliance. Schools may choose not to participate in the evaluation at all, as the program is not a regulatory requirement. Noncompliance occurs when schools join the study but deviate from the assigned program directions, such as implementing the program partially or not at all. Attrition may occur when students move out of the school district or stop completing assessments during the program, and students who move into the district partway through will lack baseline data, which can affect the data collected. Standardized implementation requirements, training, and quarterly participation audits are needed to mitigate noncompliance in the design phase. Documentation on differences in implementation will need to be tracked for clean data assessment. Randomization will occur at the school level, but data will be collected from individual students, and the analysis will keep the student-level data while adjusting standard errors for clustering of students within schools (Gertler et al., 2016). Statistical power depends mainly on the number of schools rather than the number of students, because students in the same school tend to resemble one another (Gertler et al., 2016). Collecting student-level data yields a larger total sample than school-level averages alone, and with this being a national program, a substantial number of schools may still be included, though fewer than would be available if randomization were at the student level.

Outcome Measurement

The primary outcome of this program is increased knowledge of postsecondary education options for rural high school students, with a secondary outcome of increased confidence in navigating available postsecondary options. The primary outcome variable will be the change in each student's postsecondary knowledge score from baseline to the first-year assessment, compared between treatment and comparison schools.

The evaluation team will develop a standardized multiple-choice assessment, drawing on existing postsecondary knowledge and readiness measures where possible. Approximately 20 questions will cover five areas: postsecondary pathways, admission requirements, financial aid and FAFSA, identifying programs aligned with students' goals, and finding reliable information. Each question will be scored as correct or incorrect, producing an overall knowledge score. Questions will use short scenarios, such as: "A student wants to apply for federal financial aid. What should they do first?" (a) Create an account at studentaid.gov and complete the FAFSA; (b) Wait for a college to send an application; (c) apply for scholarships online; (d) ask the school to apply on the student's behalf. To establish content validity, the evaluation team will ask school counselors and postsecondary advising experts to review whether the questions adequately represent the five knowledge areas. Items will be developed independently of the AI platform's wording, pilot-tested, and checked for internal consistency, with revisions made as needed. Equivalent versions of the assessment will be developed for baseline and follow-up, covering the same knowledge areas and difficulty levels. Pilot testing with students outside the evaluation sample will help identify unclear questions and assess whether the test is appropriate for different grade levels, as well as to assess comparability between versions, reducing the likelihood that improved scores reflect familiarity with repeated questions rather than increased knowledge.

A confidence section within the same survey will use items adapted from the College-Going Self-Efficacy Scale Short Form (Hardin et al., 2020). For example, students will rate "I know how to apply for the FAFSA" from 1 (not at all confident) to 5 (very confident), with the average response forming the confidence score.

The primary outcome is specific to postsecondary knowledge and meets the SMART criteria because it measures a specific knowledge gap through a standardized, scored assessment. Knowledge improvement is achievable through the advising activities and directly relevant to the program's goal of improving access to postsecondary information. The outcome is time-bound through the baseline and first-year assessments, with annual follow-ups used to examine whether improvements persist.

New data collection will be necessary to establish baseline confidence and knowledge scores. The evaluation team will collect new survey data using consistent procedures across treatment and comparison schools (Gertler et al., 2016). Existing administrative data from state education agencies and relevant postsecondary data providers will supplement the primary outcome with FAFSA completion, applications, and enrollment (National Student Clearinghouse, 2026). This data must cover comparable periods, include both groups, and meet completeness and reliability standards (Gertler et al., 2016). Combining new knowledge and confidence assessments with administrative data will allow the evaluation to examine whether improvements in knowledge are accompanied by changes in FAFSA completion, postsecondary applications, and enrollment, as anticipated in the Theory of Change.

Methodological Challenges & Data Needs

The evaluation requires school-level and student-level data. School-level data will establish eligibility through NCES rural classification, graduation rates, and College Readiness Index scores, as well as counselor availability, program participation, costs, and counselor workload. Student-level baseline data will include postsecondary plans, prior counselor engagement, and knowledge and confidence scores. Scores will be collected annually and one month prior to graduation to compare treatment and comparison schools, monitor attrition, and track knowledge gains. Additional outcomes include FAFSA completion, postsecondary applications, graduation rates, and postsecondary enrollment. Trained staff will administer surveys during class using school-provided devices, combining results with administrative data collected consistently across groups (Gertler et al., 2016). AI usage logs and counselor records will measure implementation and oversight.

Approximately 9.8 million students attend rural public schools nationally (National Center for Education Statistics, 2022). As of 2024, NCES classified 26,218 schools as rural (White, 2026). Eligibility will be limited to rural high schools meeting the graduation rate and College Readiness Index thresholds, verified before sampling. Schools will be stratified into NCES Rural Fringe, Distant, and Remote classifications because geographic isolation may affect access to counselors, reliable internet, and postsecondary institutions (Klasik, 2026). Stratification will support representation and comparisons across rural classifications. Schools will be randomly sampled within each stratum, then randomly assigned to implementation cohorts within the same stratum. A preliminary target of 60-90 schools, or approximately 20-30 per rural stratum, will be refined through a power calculation accounting for the minimum detectable effect, school size, and expected attrition (Gertler et al., 2016). The evaluation team will verify the number of eligible schools in each stratum before finalizing the sample.

Methodological challenges may include attrition, noncompliance, and spillover effects. Students may transfer, drop out, or stop completing assessments. Attrition rates and baseline characteristics will be compared across groups to assess potential bias, and if necessary, inverse probability weights and sensitivity analysis will be used to adjust the estimates (Gertler et al., 2016). School-level randomization reduces spillover risk, but information may spread across schools. Surveys will document cross-school exposure, and sensitivity analyses will assess possible contamination. Identical survey procedures across groups will help limit behavioral changes associated with being evaluated (Gertler et al., 2016).

Noncompliance may occur if schools fail to implement the program as intended or when comparison schools independently adopt similar AI advising tools. Platform usage logs, counselor oversight records, and annual surveys of comparison schools will document these deviations. An intent-to-treat approach will compare schools based on their original randomized assignment regardless of participation, preserving the benefits of randomization, while a complier average causal effect may be estimated using random assignment as an instrument, subject to the required assumptions (Gertler et al., 2016).

Treatment effects may differ across rural classifications due to counselor availability, reliable internet access, distance to postsecondary institutions (Klasik, 2026), or baseline postsecondary knowledge. Effects will be examined across rural classifications and the baseline characteristics described above to identify whether the program works better for particular rural populations.

Ethical & Logistical Considerations

A central ethical consideration is whether it is appropriate to provide additional advising resources to some students while delaying access for others. As this program intends to support an existing resource gap for students (Institute of Education Sciences, 2024), it is particularly important to recognize and address this concern. Because the program's effectiveness has not yet been established, and resources limit simultaneous implementation, randomized assignment through a phased rollout provides a transparent method of allocating access among eligible schools (Gertler et al., 2016). This addresses one ethical consideration of the proposed program by utilizing a transparent method of selection between eligible schools during the phased rollout.

The use of AI creates an additional ethical consideration, as AI may contain hallucinations, biased outputs, or inaccurate responses (U.S. Department of Education, Office of Educational Technology, 2023). Human oversight should remain central to this operation to minimize potential harm (U.S. Department of Education, OET, 2023). If the program lacked human oversight and students received inaccurate information related to school requirements for admission, financial aid options, or program availability, that could negatively impact the students' decisions. Due to the data being collected and the AI process involving students who are minors, there is a duty of care for privacy considerations. This duty of care involves meeting FERPA and COPPA regulations, as well as any applicable state privacy requirements (EDULEGIT Research Team, 2026). Gertler et al. (2016) emphasize protection of participants during data collection, processing, and storage, minimizing potential harm, and obtaining informed consent. Both students and their parents or guardians should be provided with detailed information in plain English, and other languages as necessary, to understand what data will be collected, including survey responses, AI platform usage logs, and counselor contact records, why, how it will be used, and who will have access to it.

The first logistical challenge I foresee includes the attempt to collect consent forms from parents or guardians. This could be due to parents with limited literacy or English proficiency skills, parents who are unavailable, or are not closely involved in their children's education (U.S. Department of Education, Office of Educational Technology, 2024). Providing materials in multiple languages and formats, such as a video, town hall format, or other community outreach options, could assist with obtaining consent. Inability to obtain sufficient consent could reduce student participation and influence participation rates between schools.

Coordinating data collection and consistent implementation across rural schools presents another logistical challenge. Accessing remote locations involves critical planning skills and proper transportation budgets, as many of these locations require significant travel to access. Successful impact evaluations require active collaboration between research and policy teams throughout the lifecycle of the evaluation (Gertler et al., 2016). Technology access may make this difficult, as rural students suffer from reduced access to high-speed internet (Lopez & Schwartz, 2019). Alternative options for students to access the internet, such as Starlink connections, mobile hotspots, and community Wi-Fi, may help to reduce this limitation (Opalka et al., 2020). Active collaboration between the research and policy teams during the program will aid in consistency of implementation, along with standardized guidance, scalable training materials, and available resources for districts to reach out to if they have questions. The additional time and resources required to ensure program adherence may help reduce implementation inconsistencies across districts.

Generalizability & Scope Conditions

Determining whether this program will work in another setting, such as an underfunded urban school, requires identifying its causal mechanisms and necessary support factors (Cartwright & Hardie, 2012). Effectiveness would depend on whether students face similar knowledge barriers and whether schools have the resources necessary to address those barriers.

Cartwright and Hardie (2012) distinguish a vertical search, which asks what causal principle makes a program work, from a horizontal search, which asks what must be in place elsewhere for that principle to operate. The Theory of Change supplies the vertical chain: personalized AI advising and counselor support improve postsecondary knowledge, which builds confidence, potentially leading to better-aligned applications, FAFSA completion, and enrollment. Because distance, transportation costs, and the desire to stay are barriers to enrollment that are stronger in rural communities than in urban ones (Klasik, 2026), a positive effect on knowledge is likely to carry over to urban schools, while the effect on enrollment could be larger or smaller, depending on which other barriers students face. Brown et al. (2016) found that low-income students in both urban and rural/suburban schools could access college information online but struggled to interpret it without knowledgeable assistance. Similarly, Shamsuddin (2024) found that students in urban high schools often had unanswered questions despite having access to online college information. These findings suggest that improving students' access to and understanding of postsecondary information may be a causal mechanism that extends beyond rural schools.

The program requires a causal team of technical support factors, including internet access, computers, and accurate AI information. These include human support factors, trained counselors and student participation, and contextual factors, local attitudes toward postsecondary education, transportation barriers, and students' preferences for remaining in their communities (Cartwright & Hardie, 2012). Sageman (2025) found that urban high schools differ in how they approach postsecondary planning; lower-resourced schools tend to have higher student-to-counselor ratios and fewer formal protocols for guiding students, while better-resourced schools provide more counselors and support. These differences may influence how students engage with the program and whether it produces similar improvements in knowledge.

The findings would be most applicable to schools meeting the original eligibility criteria: NCES Rural Fringe, Distant, or Remote schools (NCES, 2026) with graduation rates below 80% and College Readiness Index scores of 50/100 or below. Random sampling supports generalizing to eligible rural schools, but voluntary participation limits broader applicability. Reporting intent-to-treat and complier estimates will help distinguish assignment effects. Underfunded urban schools experiencing comparable advising gaps may benefit from the program, provided they have sufficient technology, counselor support, and opportunities for students to participate.

Additional research could help establish which causal mechanisms and supporting factors are most important. Pre-specified moderator analyses will examine whether effects vary by internet access and counselor availability, providing evidence about which support factors may influence program effectiveness. De-identified AI usage logs and counselor records would help examine how students engage with each component, while interviews with consenting participants could identify potential mechanisms. This data may suggest mechanisms but cannot establish the separate causal effects of AI advising and counselor support. Replicating the evaluation in underfunded urban schools would then test whether these findings would hold in a new setting.

Dissemination Plan & Political Analysis

The primary beneficiaries of expanding the AI-supported postsecondary advising program would be rural high school students with limited access to individualized college and career counseling and their families, who gain information to support students in making informed decisions. School counselors may support expansion if the program reduces repetitive advising tasks, allowing more time for individualized assistance (Stumpf, 2024). Superintendents are likely supporters if cost per student is modest but are constrained by budget (Clinton et al., 2026).

On the flip side, counselors and teachers may oppose expansion if it increases their workload or threatens to replace human guidance. Implementation costs may also create opposition among administrators facing limited budgets and competing educational priorities. School boards would respond to concerns from parents, who may express concerns about student privacy, inaccurate AI-generated information, and excessive reliance on technology (U.S. Department of Education, Office of Educational Technology, 2023).

Local school boards and district superintendents would be the primary implementation decision-makers, while state education agencies and legislators influence funding and adoption and are likely to support expansion if evidence and cost are favorable (Viennet & Pont, 2017). District leaders may support expansion if benefits justify staffing and budget demands, although competing educational priorities may favor other investments (Clinton et al., 2026). Identifying influential supporters and cultivating supportive allies can help address opposition (Peltzman & Rodriguez, 2013). Engagement should prioritize participating superintendents, school board members, and community stakeholders who could advocate for expansion.

Evaluation findings would first be presented to participating school administrators, counselors, and program staff to review implementation challenges and discuss recommendations. These participants would then serve as trusted messengers, presenting results to peers at state rural education meetings, with dissemination timed ahead of district and state budget decisions. A concise plain-language policy brief with charts would then be developed for school boards, superintendents, state education agencies, and potential funders, with the full evaluation report available on request. The brief would highlight improvements in postsecondary knowledge relative to comparison schools, FAFSA completion, postsecondary enrollment, differences across rural classifications, cost per student, and recommendations for expansion. This approach reflects the importance of tailoring findings to different audiences and sequencing dissemination to maximize policy impact (Gertler et al., 2016).

Accessible summaries would then be shared with students, families, and community organizations through school websites, newsletters, and town halls. Concerns about privacy would be answered with consent and FERPA/COPPA procedures; about AI accuracy, with counselor audit results; about workload, with counselor time data; and about cost, with cost per student. Materials would explain the findings, privacy protections, and continued role of counselors. Stakeholder feedback would be solicited to identify concerns affecting participation and local support (Gertler et al., 2016).

A phased expansion into additional eligible rural districts is recommended, prioritizing schools with significant advising gaps and sufficient implementation capacity. Monitoring of participation, student knowledge, costs, and counselor workload on a continuous basis would help inform future funding decisions and address concerns about program sustainability.

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